The inventory floor needs the 2-cell datum: $D1$ versus $D2$

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Abstract

Abstracting a supply chain to a small category $\mathcal{C}$ with fungible stock glued up to a relabelling group $K = \mathrm{Aut}F$, the obstruction to a *strict global inventory* is a K-gerbe over the nerve $N(\mathcal{C})$, staged following Giraud: a nonabelian band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ must vanish before an abelian class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ is defined. The "floor corollary" claimed that over the band-trivial stratum a strict global inventory exists exactly when $[\mathfrak{g}] = 0$. This is false, and the reason is instructive: the claim silently fixes a notion of "strict" that is one cohomological degree below $[\mathfrak{g}]$. Throughout we take the transport to be a *strict* functor (trivial coherence datum $k \equiv e$): this is the regime in which the band-triviality criterion applies and the dichotomy is clean. We separate the two honest notions. Retaining the coherence (2-cell) datum, a $D2$ global inventory exists iff $[\mathfrak{g}] = 0$ in $H^2(N(\mathcal{C}); Z(K))$; discarding it, a $D1$ global inventory exists iff a degree-one class $o_{D1} \in H^1(\mathcal{C}; \mathrm{Inn}K)/\mathrm{im}\, H^0(\mathcal{C}; \mathrm{Out}K)$ vanishes. The $D1$ condition implies the $D2$ condition ($o_{D1} = 0 \Rightarrow [\mathfrak{g}] = 0$), and the implication is strict; the converse $D2 \Rightarrow D1$ is *false*. Already over a one-dimensional base, where $H^2(N(\mathcal{C}); Z(K)) = 0$ forces $[\mathfrak{g}] = 0$ for every K, the class o_{D1} can still detect a nontrivial inner transport. The witness is $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $K = S_3$, where $\beta = 0$ and $[\mathfrak{g}] = 0$ yet no naive strictification exists; its finite core is machine-checked, and an independent sweep of 248 strict-α cases corroborates the dichotomy with the designed controls failing as they must.

1 Introduction

The floor, and a false corollary. A supply chain is a small category $\mathcal{C}$: objects are nodes, morphisms are logistics operations, composition is sequencing. When local inventory charts are comparable only *up to a relabelling* of stock states – up to the automorphism group $K := \mathrm{Aut}F$ of the stock fibre F – the gluing datum is an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$, and the obstruction to a *strict global inventory* (a single coherent trivialization across the chain) is staged in the manner of Giraud [3]: a nonabelian band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ must vanish before an abelian gerbe class

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$[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ is even defined. On the band-trivial stratum $\beta = 0$ it is tempting to close the account with a clean biconditional:

$$\text{(floor corollary, } \textit{false}\text{)} \qquad \text{strict global inventory exists} \iff \beta = 0 \ \wedge \ [\mathfrak{g}] = 0.$$

It is false. Rick, refereeing the band-boundary note, recomputed the obstruction for $\mathcal{C} = \mathrm{B}(\mathbb{Z})$ with $K = S_3$ and found $\beta = 0$ and $[\mathfrak{g}] = 0$ yet no strictification: the reflection transport cannot be relabelled to the identity. This note explains *why*, and in doing so turns the corollary into a pair of correct theorems.

Standing hypothesis (strict transport). Throughout this note the transport $\alpha\colon \mathcal{C} \to \mathrm{B}(\mathrm{Aut}K)$ is a *strict* functor: its coherence datum is trivial, $k_{g,f} \equiv e$. This is the regime in which the band-triviality criterion (clause (a), §2) applies verbatim, so that $\beta = 0$ yields a strict inner transport $\bar{\alpha}$, and in which Definition (D2) below may omit the comparison datum. For a genuinely non-strict α (nontrivial $k_{g,f}$) neither holds as printed: the band-trivial stratum need not yield a strict $\bar{\alpha}$ even when $\beta = 0$ and $[\mathfrak{g}] = 0$ (already for $\mathcal{C} = \mathrm{B}(\mathbb{Z}/3)$, or $K = D_4$), and both notions must then carry the k-datum. That extension is deliberately outside the present scope; everything below is stated under $k_{g,f} \equiv e$.

The gap: two meanings of "strict". The corollary hides a choice. Once the band is trivial, the transport is gauge-equivalent to a strict *inner* functor $\bar{\alpha}\colon \mathcal{C} \to \mathrm{B}(\mathrm{Inn}K)$ (Section 2); the question is what we ask of it. Two answers:

(D1) *(naive)* a vertex gauge $\mu\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ relabelling the transport away *entirely*: $\mu(y)\,\bar{\alpha}(f)\,\mu(x)^{-1} = \mathrm{id}_K$ for every $f\colon x \to y$. No coherence datum is retained.

(D2) a gauge μ *together with* a strict K-valued functor $T\colon \mathcal{C} \to \mathrm{B}K$ ($T_{gf} = T_g T_f$) realising the transport by honest group elements: $\mu(y)\,\bar{\alpha}(f)\,\mu(x)^{-1} = c_{T_f}$, where $c_a = \mathrm{conj}(a)$. (Under the standing strict-transport hypothesis $\bar{\alpha}$ is strict, so this equation carries no coherence term; for non-strict α it must be written with the comparison datum $k_{g,f}$, and then T need not be strict — exactly the extension we exclude.)

$D1$ demands that the transport be *trivialised*; $D2$ demands only that it *lift* to the group K, surviving as a genuine inner relabelling carried by elements of K. These are different requests, and – this is the whole point – they live one cohomological degree apart.

The result. Write $Z = Z(K)$, $\mathrm{Inn}K = K/Z$, $\mathrm{Out}K = \mathrm{Aut}K/\mathrm{Inn}K$, $q\colon \mathrm{Aut}K \twoheadrightarrow \mathrm{Out}K$, and let $[\mathfrak{g}] := \bar{\alpha}^*[\star] \in H^2(N(\mathcal{C}); Z)$ be the pullback along $\bar{\alpha}$ of the class $[\star]$ of the central extension $1 \to Z \to K \to \mathrm{Inn}K \to 1$ (Section 2; this is the gerbe class of [4]).

Theorem 1 (Floor dichotomy)**.** *Fix a small category $\mathcal{C}$ with band-trivial transport $\bar{\alpha}\colon \mathcal{C} \to \mathrm{B}(\mathrm{Inn}K)$.*

1. *(D2* ***end.****) A D2 global inventory exists if and only if $[\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z)$.*
2. *(D1* ***end.****) A D1 global inventory exists if and only if $o_{D1} = 0$, where o_{D1} is the class of $\bar{\alpha}$ in the pointed quotient $H^1(\mathcal{C}; \mathrm{Inn}K)/\mathrm{im}\, H^0(\mathcal{C}; \mathrm{Out}K)$. This is a* degree-one *obstruction; it is not $[\mathfrak{g}]$.*
3. *(D1 $\Rightarrow$ D2,* ***strictly.****) $o_{D1} = 0 \Rightarrow [\mathfrak{g}] = 0$, and the implication is strict. The converse D2 $\Rightarrow$ D1 fails: already over a one-dimensional base, where $H^2(N(\mathcal{C}); Z) = 0$ forces $[\mathfrak{g}] = 0$*

for every K, the class o_{D1} can be nonzero (part 4). Centre size is not the governing parameter — the gap occurs both over centreless stock ($K = S_3$) and over stock with nontrivial centre ($K = D_4$, $Z = \mathbb{Z}/2$).

4. ***(Separation.)*** *For $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $K = S_3$, $\alpha = c_{(0\,1)}$: $\beta = 0$ and $[\mathfrak{g}] = 0$ (so $D2$ holds – an explicit lift is $T(n) = \sigma^n$), yet $o_{D1} \neq 0$ (no naive gauge strictifies the reflection). Hence the floor corollary is false, and the honest $D1$ floor lives one degree down.*

The method. One slogan runs through the proof: *name the extension, and its kernel names both the coefficient group and the cohomological degree.* $D2$ asks the inner transport to lift through the central extension ($\star$) with kernel Z; the lifting obstruction is a degree-two class with coefficients in Z – exactly $[\mathfrak{g}]$ (Lemma 2, the Eilenberg–Mac Lane / Brown criterion [2, 1]). $D1$ asks the transport to be a *coboundary* in the nonabelian $H^1(\mathcal{C}; \mathrm{Inn}K)$ – a degree-one condition with coefficients in $\mathrm{Inn}K$. Discarding the coherence datum costs precisely one degree, and shifts the coefficient group from the centre Z up to the inner group $\mathrm{Inn}K$. Keeping these two coefficient groups distinct is the entire content of the repair.

Consequences. Over a one-dimensional base the floor is always clear on the $D2$ side: $N(\mathrm{B}(\mathbb{Z})) \simeq S^1$ has $H^2 = 0$, so a $D2$ inventory exists for every K (Corollary 4). The $D1$ failure for S_3 is therefore not a failure of the gerbe class at all – it is the degree-one shadow the corollary could not see. For the grant's Applications pillar the moral is operational: whether a supply chain admits a coherent global relabelling depends on *which* notion of "global relabelling" the operator can afford. If the stock is genuinely fungible (centreless relabelling group), demanding the strong form $D1$ can fail even when the weak form $D2$ – carry the relabelling as an explicit, composable group-valued correction – succeeds.

Outline. Section 2 fixes the extension ($\star$), the nonabelian H^1, and the band-trivialised transport. Section 3 proves the $D2$ end (Theorem 1.1). Section 4 constructs o_{D1} and proves the $D1$ end (1.2). Section 5 records the degree lane and $D1 \Rightarrow D2$ (1.3). Section 6 works the S_3 separation (1.4) and cites its machine-checked core.

2 Preliminaries

We recall only what the proof uses. Let $K = \mathrm{Aut}F$ be the relabelling group of the stock fibre, $Z = Z(K)$ its centre, $\mathrm{Inn}K = K/Z$ the inner automorphism group, $\mathrm{Out}K = \mathrm{Aut}K/\mathrm{Inn}K$ the outer group; $q\colon \mathrm{Aut}K \twoheadrightarrow \mathrm{Out}K$ is the projection and $c\colon K \twoheadrightarrow \mathrm{Inn}K \subseteq \mathrm{Aut}K$, $c_a = \mathrm{conj}(a)$, is conjugation.

The central extension. Conjugation gives a central extension of groups

$$1 \longrightarrow Z \longrightarrow K \xrightarrow{c} \mathrm{Inn}K \longrightarrow 1, \qquad [\star] \in H^2(\mathrm{Inn}K; Z), \tag{$\star$}$$

central because inner automorphisms fix Z pointwise, so the action is trivial.

The transport and its band. The inventory datum is an *outer transport*: a pseudofunctor $\mathcal{C} \to \mathrm{B}(\mathrm{Aut}K)$ sending $f\colon x \to y$ to $\alpha(f) \in \mathrm{Aut}K$ with coherence (2-cell) data $k_{g,f} \in K$, $\alpha(gf) = c_{k_{g,f}}\,\alpha(g)\alpha(f)$. Its *band* is the strict functor $\rho = q \circ \alpha\colon \mathcal{C} \to \mathrm{B}(\mathrm{Out}K)$, with class $\beta = [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ (nonabelian H^1). Under the standing hypothesis (§1) α is *strict*, $k_{g,f} \equiv e$: an honest functor $\mathcal{C} \to \mathrm{B}(\mathrm{Aut}K)$, with ρ its band.

Standing hypothesis: the band is trivial. Throughout, $\beta = 0$. By the band-triviality criterion (clause (a) of the band-boundary note, computed) *for strict* α, the functor α is $\mathrm{Aut}K$-gauge-equivalent to a *strict* $\mathrm{Inn}K$-valued functor. (Clause (a) is stated and verified for strict α; for non-strict α it can fail even at $\beta = 0$, which is precisely why strictness is a standing hypothesis.) Fix such a gauge once and write $\bar{\alpha}\colon \mathcal{C} \to \mathrm{B}(\mathrm{Inn}K)$, $\bar{\alpha}(gf) = \bar{\alpha}(g)\bar{\alpha}(f)$, $\bar{\alpha}(\mathrm{id}) = \mathrm{id}$, for the band-trivialised transport. This is the object both $D1$ and $D2$ act on.

Nonabelian H^1. For a group G with trivial $\mathcal{C}$-action, let $H^1(\mathcal{C}; G)$ be the pointed set of functors $\mathcal{C} \to \mathrm{B}G$ modulo natural isomorphism (Serre [6, I.5]). Concretely a functor is a 1-*cocycle* $f \mapsto g_f \in G$ with $g_{\mathrm{id}} = e$ and $g_{gf} = g_g g_f$; a natural isomorphism is *vertex data* $\nu\colon \mathrm{Ob}\mathcal{C} \to G$ acting by $g_f \mapsto \nu(y) g_f \nu(x)^{-1}$. The basepoint $*$ is the trivial cocycle $f \mapsto e$. A cocycle is a *coboundary* (equivalent to $*$) iff $g_f = \nu(y)^{-1}\nu(x)$ for some ν. For G nonabelian this is only a pointed set; that is all we need. The transport $\bar{\alpha}$ is such a 1-cocycle valued in $\mathrm{Inn}K$; for $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $H^1(\mathrm{B}(\mathbb{Z}); G) = G/\mathrm{conj}$ and $[\bar{\alpha}]$ is the conjugacy class of $\bar{\alpha}(1)$.

The gerbe class. Define $[\mathfrak{g}] := \bar{\alpha}^*[\star] \in H^2(N(\mathcal{C}); Z)$, the pullback of the extension class ($\star$) along the strict functor $\bar{\alpha}\colon N(\mathcal{C}) \to \mathrm{B}(\mathrm{Inn}K)$. That this is the $\mathrm{Aut}F$-gerbe's obstruction, and is independent of the coherence datum, is the content of the capstone's rectification theorem [4]; we take it as given.

3 The $D2$ floor: the obstruction is $[\mathfrak{g}] \in H^2$

The key idea is that $D2$ is exactly a *strict lifting problem* through the central extension ($\star$), and such problems are governed by a degree-two class with coefficients in the kernel Z.

Lemma 2 (Strict lifting through a central extension)**.** *Let $\varphi\colon \mathcal{C} \to \mathrm{B}(\mathrm{Inn}K)$ be a strict functor. A strict functor $T\colon \mathcal{C} \to \mathrm{B}K$ with $c \circ T = \varphi$ (that is, $T_f \in K$, $c(T_f) = \varphi_f$, $T_{gf} = T_g T_f$) exists if and only if $\varphi^*[\star] = 0 \in H^2(N(\mathcal{C}); Z)$.*

Proof. Choose a set-section $s\colon \mathrm{Inn}K \to K$ of c with $s(e) = e$ and put $t_f := s(\varphi_f)$, so $c(t_f) = \varphi_f$. The defect $z(g, f) := t_{gf}(t_g t_f)^{-1}$ satisfies $c(z(g, f)) = \varphi_{gf}(\varphi_g \varphi_f)^{-1} = e$, hence $z(g, f) \in Z$. Associativity of composition in $\mathcal{C}$ forces the 2-cocycle identity $\delta z = 0$ on the nerve $N(\mathcal{C})$ (the standard bar computation; Z central, so the action is trivial), and $[z] \in H^2(N(\mathcal{C}); Z)$ is exactly $\varphi^*[\star]$ – the pullback of the bar 2-cocycle of ($\star$) along φ [1, IV.3], [2]. A strict lift is a choice $T_f = t_f \zeta_f$ with $\zeta_f \in Z$ making the defect vanish; since $z' = z \cdot \delta\zeta$, a strict lift exists iff z is a Z-valued coboundary iff $[z] = 0$. □

Proposition 3 ($D2$ end)**.** *A $D2$ global inventory exists if and only if $[\mathfrak{g}] = \bar{\alpha}^*[\star] = 0 \in H^2(N(\mathcal{C}); Z)$.*

Proof. Given (μ, T), set $\tau(f) := \mu(y)\, \bar{\alpha}(f)\, \mu(x)^{-1}$. Since $\mathrm{Inn}K \trianglelefteq \mathrm{Aut}K$ and $\bar{\alpha}(f) \in \mathrm{Inn}K$, $\tau(f)$ is inner, and strictness of $\bar{\alpha}$ gives $\tau(gf) = \mu(z)\bar{\alpha}(gf)\mu(x)^{-1} = \tau(g)\tau(f)$, so $\tau\colon \mathcal{C} \to \mathrm{B}(\mathrm{Inn}K)$ is a *strict* functor. The $D2$ equation $\tau(f) = c_{T_f}$ with T strict says precisely that T is a strict lift of τ through ($\star$); by Lemma 2 such (μ, T) exists iff $\tau^*[\star] = 0$ for some admissible μ.

It remains to see $\tau^*[\star] = 0 \iff \bar{\alpha}^*[\star] = 0$. Decompose $\mu(x) = \hat{h}_c\, j(x)$ on each component c, where $\hat{h}_c \in \mathrm{Aut}K$ is a fixed lift of the locally constant outer part $q \circ \mu$ and $j(x) \in \mathrm{Inn}K$. Then $\tau(f) = \mathrm{Ad}(\hat{h}_c)\big(j(y)\, \bar{\alpha}(f)\, j(x)^{-1}\big) = \mathrm{Ad}(\hat{h}_c) \circ \bar{\alpha}'(f)$, where $\bar{\alpha}'(f) := j(y)\bar{\alpha}(f)j(x)^{-1}$ is naturally isomorphic to $\bar{\alpha}$ (inner vertex data), so $\bar{\alpha}'^* = \bar{\alpha}^*$ on H^2. The global automorphism $\hat{h}_c$ preserves ($\star$) (it fixes Z setwise, inducing $\hat{h}_c|_Z \in \mathrm{Aut}Z$, and covers $\mathrm{Ad}(\hat{h}_c) \in \mathrm{Aut}(\mathrm{Inn}K)$), so $\tau^*[\star] = (\hat{h}_c|_Z)_*\, \bar{\alpha}^*[\star] =$

$(\hat{h}_c|_Z)_*[\mathfrak{g}]$, an automorphism image of $[\mathfrak{g}]$. An automorphism of $H^2(N(\mathcal{C}); Z)$ fixes 0, so $\tau^*[\star] = 0 \iff [\mathfrak{g}] = 0$. □

Corollary 4. *For $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $N(\mathcal{C}) \simeq S^1$, so $H^2(S^1; Z) = 0$ and $[\mathfrak{g}] = 0$ for every K: a $D2$ global inventory always exists over a free one-generator base.*

4 The $D1$ floor: the obstruction drops to $o_{D1} \in H^1$

The key idea here is dual to the last: $D1$ asks $\bar{\alpha}$ not to *lift* but to *vanish* – to be a coboundary in the nonabelian $H^1(\mathcal{C}; \mathrm{Inn}K)$. This is a degree-one condition, and its coefficient group is $\mathrm{Inn}K$, not Z.

Band-preservation is forced. If a $D1$ gauge μ exists, applying q to $\mu(y)\bar{\alpha}(f)\mu(x)^{-1} = \mathrm{id}$ and using $q(\bar{\alpha}(f)) = e$ (band trivial) gives $q(\mu(y)) = q(\mu(x))$ for every $f\colon x \to y$. Hence $q \circ \mu\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Out}K$ is locally constant: $D1$ *entails* band-preservation, with no extra hypothesis.

Construction of o_{D1}. $H^0(\mathcal{C}; \mathrm{Out}K)$ is the group of locally constant maps $\mathrm{Ob}\mathcal{C} \to \mathrm{Out}K$ ($= \mathrm{Out}K$ per component). It acts on the pointed set $H^1(\mathcal{C}; \mathrm{Inn}K)$: given $\theta \in H^0(\mathcal{C}; \mathrm{Out}K)$, choose a locally constant lift $\hat{h}\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ with $q \circ \hat{h} = \theta$, and send a cocycle g_f to $\hat{h}(y)\, g_f\, \hat{h}(x)^{-1} \in \mathrm{Inn}K$ (inner, as $\mathrm{Inn}K \trianglelefteq \mathrm{Aut}K$). A different lift of θ differs by an $\mathrm{Inn}K$-valued cochain, changing the result by an $\mathrm{Inn}K$-coboundary; so the action descends to $H^1(\mathcal{C}; \mathrm{Inn}K)$ and fixes the basepoint $*$. Set

$$o_{D1} := [\bar{\alpha}] \;\in\; H^1(\mathcal{C}; \mathrm{Inn}K) \,/\, \mathrm{im}\, H^0(\mathcal{C}; \mathrm{Out}K).$$

Proposition 5 ($D1$ end)**.** *A $D1$ global inventory exists if and only if $o_{D1} = 0$; equivalently, iff $[\bar{\alpha}] = *$ in $H^1(\mathcal{C}; \mathrm{Inn}K)$.*

Proof. We first show $D1 \iff [\bar{\alpha}] = *$.

- If $[\bar{\alpha}] = *$, pick $j\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Inn}K$ with $\bar{\alpha}(f) = j(y)^{-1}j(x)$. Set $\mu := j$. Then $\mu(y)\bar{\alpha}(f)\mu(x)^{-1} = j(y)\, j(y)^{-1}j(x)\, j(x)^{-1} = \mathrm{id}$, so $D1$ holds.
- Conversely let $\mu\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ solve $D1$. By the forced constraint $q \circ \mu = \theta_c$ is constant on each component c; choose a lift $\hat{h}_c \in \mathrm{Aut}K$ of θ_c and write $\mu(x) = \hat{h}_c\, j(x)$ with $j(x) := \hat{h}_c^{-1}\mu(x) \in \ker q = \mathrm{Inn}K$. For $f\colon x \to y$ in c, $\bar{\alpha}(f) = \mu(y)^{-1}\mu(x) = j(y)^{-1}j(x)$, an $\mathrm{Inn}K$-coboundary on c. Over all components, $[\bar{\alpha}] = *$.

Now the quotient. The $\mathrm{Out}K$-action fixes $*$, so the fibre of the quotient map over the basepoint is exactly the $\mathrm{Out}K$-orbit of $*$, namely $\{*\}$. Hence $o_{D1} = 0$ (the basepoint of the quotient) iff $[\bar{\alpha}] = *$ iff $D1$ holds. □

The quotient by $\mathrm{im}\, H^0(\mathcal{C}; \mathrm{Out}K)$ makes o_{D1} independent of the choice of fibre coordinates (a global automorphism of K acts by an outer class). The action can be genuinely nontrivial – for $K = D_4$ it swaps the two reflection conjugacy classes, both nonzero – so the quotient is needed for coordinate-freeness, but by the last line of the proof it never changes *whether* $o_{D1} = 0$: the quotient is vanishing-neutral.

5 The degree lane, and $D1 \Rightarrow D2$

Remark 6 (one well-definedness caveat, and the degree drop)**.** Both invariants are canonical only up to the residual Out$K$-action (global fibre re-coordinatisation): $[\mathfrak{g}]$ up to the Out$K$-action on $H^2(N(\mathcal{C}); Z)$ (Proposition 3), o_{D1} by the explicit quotient (Proposition 5). In both cases the action is by automorphisms fixing 0 (resp. $*$), so vanishing is invariant and both biconditionals are well-posed. $D1$ and $D2$ differ not in their gauge symmetry but in their *degree*: $D1$ asks $\bar{\alpha}$ to be a coboundary – trivialised to the identity, a degree-one (H^1) condition on Inn$K$; $D2$ asks $\bar{\alpha}$ only to lift to a strict K-valued functor, a degree-two (H^2, k-invariant) condition on Z. Dropping the coherence datum costs exactly one cohomological degree and raises the coefficient group from Z to Inn$K$.

Proposition 7 ($D1 \Rightarrow D2$, strictly)**.** $o_{D1} = 0 \Rightarrow [\mathfrak{g}] = 0$*, and the containment is strict.*

Proof. If $[\bar{\alpha}] = *$ then $\bar{\alpha}$ is naturally isomorphic to the constant functor at the identity object of B(Inn$K$); pullback along a functor isomorphic to a constant kills H^2, so $[\mathfrak{g}] = \bar{\alpha}^*[\star] = 0$ (equivalently, take $T \equiv e$ in $D2$). Strictness: the converse $D2 \Rightarrow D1$ fails already over a one-dimensional base, where $H^2(N(\mathcal{C}); Z) = 0$ forces $[\mathfrak{g}] = 0$ for every K while $\bar{\alpha}$ may be a nontrivial inner transport. $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $K = S_3$ (§6) is such a witness, with $Z = 1$; so is $K = D_4$ with $\bar{\alpha} = c_s$ a reflection, where $Z = \mathbb{Z}/2 \neq 1$ — so centre size does not govern the gap, and the earlier "Z too small" heuristic is not a characterisation. □

Remark 8 (what the gap is, and is not)**.** The gap $o_{D1} \neq 0 = [\mathfrak{g}]$ is a phenomenon of cohomological *degree*, not of centre size. On a base with $H^2 = 0$ the degree-two class $[\mathfrak{g}]$ is forced to vanish for *every* coefficient group, so $D2$ holds unconditionally, yet the degree-one class o_{D1} still records the inner transport. An exact description of the locus $\{\, o_{D1} \neq 0 = [\mathfrak{g}] \,\}$ over a general base is not attempted here.

6 The separation: $\mathcal{C} = \mathrm{B}(\mathbb{Z})$, $K = S_3$

S_3 is *complete*: $Z(S_3) = 1$, Inn$S_3 = S_3$, Out$S_3 = 1$, Aut$S_3 \cong S_3$ [5, Thm 7.5]. The transport on B($\mathbb{Z}$) is determined by the generator, $\alpha(1) = c_\sigma$ with $\sigma = (0\,1)$, and freely $\alpha(n) = c_{\sigma^n}$; as B($\mathbb{Z}$) is free on one generator this is already strict with trivial coherence datum, so $\bar{\alpha}(n) = c_{\sigma^n}$.

- **Band.** Out$S_3 = 1$, so $\beta \in H^1(S^1; 1) = 0$. ✓
- **Gerbe class.** $Z(S_3) = 1$, so $H^2(S^1; Z) = 0$ and $[\mathfrak{g}] = 0$; independently, $N(\mathrm{B}(\mathbb{Z})) \simeq S^1$ forces $H^2 = 0$ (Corollary 4). By Proposition 3 a $D2$ inventory exists, with explicit lift $T(n) = \sigma^n$: a homomorphism $\mathbb{Z} \to S_3$ since $\sigma^2 = e$, and $\mu = \mathrm{id}$, $c_{T(n)} = c_{\sigma^n} = \bar{\alpha}(n)$. ✓
- o_{D1}**.** $H^1(\mathrm{B}(\mathbb{Z}); \mathrm{Inn}S_3) = S_3/\mathrm{conj} = \{[e], [\text{transposition}], [\text{3-cycle}]\}$, and o_{D1} is the class of σ, a transposition $\neq [e]$; with Out $= 1$ it is left unchanged, so $o_{D1} \neq 0$. By Proposition 5 no $D1$ inventory exists. Directly: on one object $\mu\, \bar{\alpha}(1)\, \mu^{-1} = c_{m\sigma m^{-1}}$, and $c_{m\sigma m^{-1}} = \mathrm{id} \iff m\sigma m^{-1} = e$ (as $Z(S_3) = 1$ makes c injective), impossible for $\sigma \neq e$. ✓

Hence $\beta = 0 \wedge [\mathfrak{g}] = 0$ while $D1$ fails: the floor corollary is false, and the honest $D1$ floor lives one degree down in H^1. The finite, decidable core of this witness – $Z(S_3) = 1$, AutS_3 of order 6 (so Out$S_3 = 1$), the $D2$ lift $\sigma^2 = e$, and the exhaustive check that no $m \in S_3$ conjugates σ to e – is machine-checked in Lean 4 (`floor_separation`, kernel-checked with only `propext` and `Quot.sound`, no `native_decide` and no added axioms). The cohomological identifications

($[\mathfrak{g}] = \bar{\alpha}^*[\star]$, the location of o_{D1}, and the forcing of vanishing from trivial coefficient groups) are the cited classical bridges, not re-derived in Lean. Beyond this witness, an independent recomputation of the dichotomy of Theorem 1 over 248 strict-α cases (independent referee recomputation; `work-in-progress scripts/day238`) found 0 violations, with the designed controls failing exactly as they must.

Upshot

Rick's $B(\mathbb{Z})/S_3$ counterexample is now a theorem. The inventory floor splits cleanly by *how much structure one retains*. Keep the coherence (2-cell) datum ($D2$): the obstruction is the gerbe class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$. Discard it ($D1$): the obstruction drops a degree to $o_{D1} \in H^1(\mathcal{C}; \mathrm{Inn}K)/\mathrm{im}\, H^0(\mathcal{C}; \mathrm{Out}K)$. The $D1$ condition implies the $D2$ condition, strictly; the converse fails already over a one-dimensional base, where $H^2 = 0$ makes $[\mathfrak{g}]$ vanish for every K while o_{D1} still sees a nontrivial inner transport. Both $\mathrm{B}(\mathbb{Z})/S_3$ (centreless) and $\mathrm{B}(\mathbb{Z})/D_4$ (centre $\mathbb{Z}/2$) are honest witnesses: the gap is a matter of cohomological degree, not of centre size. The repair isolates the single recurring hazard in the program – confusing the central coefficient $Z(K)$ of the degree-two class with the inner coefficient $\mathrm{Inn}K$ of the degree-one one. Whether this re-promotes the band-boundary note is a separate registry judgement, deferred; here we have stated only what is proved. The scope is the band-trivial stratum $\beta = 0$; the nontrivial-band and T^2-coupled frontiers are treated in the companion capstone [4].

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